

Mahindra University Hyderabad
École Centrale School of Engineering
Supplementary Examination

Program: B. Tech.

Branch: CM
Subject: Signals and Systems (MA2105)

Year: II

Semester: I

Date: 12/09/2025

Time Duration: 1.5 Hours

Start Time: 10:00 AM

Max. Marks: 15

Instructions:

- 1) Answer all the questions.
- 2) All questions are self-explanatory; no clarification will be provided during the exam.

1. (3 points) Consider the triangular signal

$$x(t) = \begin{cases} 0, & t \leq -1, \\ 1+t, & -1 < t \leq 0, \\ 1-t, & 0 < t < 1, \\ 0, & t \geq 1. \end{cases}$$

- (a) Find an explicit expression for $y_1(t) = x(2t)$ and plot it.
 - (b) Find an explicit expression for $y_2(t) = x(0.5t)$ and plot it.
 - (c) Find the explicit expression for $y_3(t) = x(t-1)$ and plot it.
2. (4 points) Consider a continuous-time Linear Time Invariant (LTI) system with impulse response

$$h(t) = e^{-2t}u(t),$$

where $u(t)$ is the unit step signal. Determine the following properties of the system:

- (a) Does the system have memory, or is it memoryless?
- (b) Is the system causal or noncausal?
- (c) Is the system Bounded-Input Bounded-Output (BIBO) stable?

or

Consider a discrete-time LTI system with impulse response

$$h[n] = 3^n u[-n] + \left(\frac{1}{3}\right)^n u[n],$$

where $u[n]$ is the unit step signal. Determine the following properties of the system:

- (a) Does the system have memory, or is it memoryless?
- (b) Is the system causal or noncausal?
- (c) Is the system BIBO stable?

3. (4 points) Determine whether the continuous-time signal

$$x(t) = e^{-t}u(t),$$

is an energy signal, a power signal, or neither. Here $u(t)$ denotes the unit step signal.

or

Determine whether the discrete-time signal

$$x[n] = \left(\frac{1}{4}\right)^n u[n],$$

is an energy signal, a power signal, or neither. Here $u[n]$ denotes the unit step signal.

4. (4 points) Do the following questions:

(a) Find the fundamental period of the signal

$$x[n] = \sin\left(\frac{2\pi}{7}n\right) + \cos\left(\frac{\pi}{2}n\right).$$

(b) Compute

$$\int_{-\infty}^{\infty} e^{-\tau} \delta(\tau - 4) d\tau.$$

(c) Determine whether the system H , defined by

$$H\{x(t)\} = 13[x(t) - 2],$$

is linear or not.

(d) Consider the piecewise signal with intervals on both sides of the origin

$$x(t) = \begin{cases} t + 3, & -3 \leq t < -1, \\ 0, & -1 \leq t < 0, \\ 4 - 2t, & 0 \leq t < 2, \\ (t - 2)^3, & 2 \leq t < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Express $x(t)$ as a single formula using $u(t)$, where $u(t)$ is the unit step signal.